

CFR-MARL: Centralized feedback-driven reward multi-agent reinforcement learning for decentralized cooperative path planning of heterogeneous agents

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ABSTRACT

Coordinating heterogeneous robot teams for efficient, full-area coverage in unknown environments demands robust, fully autonomous control. Outer-planet exploration such as missions to Mars exemplifies this challenge, requiring reliable autonomy to guarantee mission success. To overcome the difficulties of decentralized, heterogeneous multi-agent coordination, this paper presents Centralized Feedback-Driven Reward Multi-Agent Reinforcement Learning (CFR-MARL), a decentralized multi-agent reinforcement learning framework designed for cooperative coverage path planning in heterogeneous teams of unmanned ground vehicles (UGVs) and an unmanned aerial vehicle (UAV). The CFR-MARL integrates a Value Decomposition Network (VDN) for centralized training with potential-based reward shaping, delivering dense, team-level feedback while enabling fully decentralized execution with minimal communication overhead. Through a shared discretized map, the UAV performs real-time obstacle detection, coverage validation, and energy logistics, dynamically guiding UGVs to unexplored areas and preventing redundant traversal. Simulation results demonstrate that CFR-MARL substantially accelerates learning, reduces coverage overlap, and nearly eliminates mission failures due to collisions or energy depletion, while scaling effectively to larger UGV teams. These results establish the CFR-MARL as an efficient and scalable solution for resource-constrained planetary exploration and other heterogeneous robotic applications.

Acronym	Full Form
CFR-MARL	Centralized Feedback-Driven Reward Multi-Agent Reinforcement Learning
CPP	Coverage Path Planning
CR	Collision Rate
DDPG	Deep Deterministic Policy Gradient
DM	Discretized Communication Data Matrix
DQN	Deep Q-Network
FOV	Field of View
IMU	Inertial Measurement Unit
MADDPG	Multi-Agent Deep Deterministic Policy Gradient
MARL	Multi-Agent Reinforcement Learning
MAS	Multi-Agent Systems
MDP	Markov Decision Process
OR	Overlap Ratio
QMIX	Monotonic value-function factorization for deep multi-agent RL
RL	Reinforcement Learning
SR	Success Rate

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TC	Total Coverage Time
UAV	Unmanned Aerial Vehicle
UGV	Unmanned Ground Vehicle
VDN	Value Decomposition Network

1. Introduction

Heterogeneous multi-agent systems (MAS) consist of robots with differing dynamics, sensors, and capabilities collaborating to achieve a common objective. Such teams might include ground vehicles specialized for rough-terrain traversal, aerial vehicles capable of rapid, wide-area sensing, and even aquatic or climbing robots for 3D environments [1]. By leveraging each platform's unique strengths, heterogeneous MAS can cover complex environments more quickly, tolerate individual

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failures, and adapt to diverse mission requirements. However, these benefits come with significant challenges. Agents have disparate state and action spaces, making joint policy design and credit assignment nontrivial; communication constraints and asynchronous sensing complicate information sharing; and differences in mobility and energy budgets demand coordinated task allocation to avoid bottlenecks or idle resources. In the domain of autonomous path planning and coverage path planning (CPP), these issues translate into the need for algorithms that respect each agent's kinematics and sensing footprint while ensuring global coverage.

A heterogeneous robotic team can combine unmanned ground vehicles (UGVs), outfitted with rugged drivetrains and ground-penetrating sensors for close-range terrain inspection, with unmanned aerial vehicles (UAVs) that carry LiDAR and high-resolution cameras for rapid overhead surveys [2]. UGVs are capable of negotiating steep inclines and narrow crevices but lack the long-range sensing needed to detect distant hazards [3], whereas UAVs excel at wide-area mapping yet are constrained by limited flight endurance and cannot directly manipulate ground obstacles [4]. An effective CPP framework must therefore assign reconnaissance flights to the UAVs for continuously updating a shared discretized map and validating safe corridors, while guiding the UGVs through those corridors to achieve detailed, fine-grained coverage [5]. At the same time, the system must coordinate UAV landings on UGV charging stations for battery replenishment and data transfer, avoiding redundant traversal throughout the mission [6]. All of these tasks must be accomplished through a decentralized, scalable coordination scheme in which each agent bases its decisions on local observations augmented only by intermittently synchronized map information, thus ensuring full-area coverage, minimal overlap, and resilience to communication bottlenecks. Experimental studies have also highlighted the benefits and practical feasibility of aerial-ground cooperation on physical robotic platforms. Fankhauser et al. [7] demonstrate collaborative navigation between a flying and a walking robot in challenging terrain, and emphasize system-level integration of perception, localization, and motion control. Khabbaz and Nokleby [8] investigate a UAV-assisted obstacle-mapping framework for multi-UGV exploration and mapping, in which aerial observations are incorporated into the ground robot's cost maps. These studies show that tightly coupled UAV–UGV teams can enhance perception and navigation in real environments and provide valuable insights into the design of heterogeneous robotic systems.

Mars exploration is a critical endeavor in planetary science, with numerous missions aimed at mapping the planet's surface, studying its geology, and searching for signs of past or present life [9]. NASA's Perseverance mission and the European Space Agency's ExoMars project represent key examples of efforts aimed at exploring Mars' surface and subsurface for biosignatures [10]. Given the challenges posed by the harsh and unpredictable Martian environment, autonomous robotic systems play an indispensable role in these missions [11]. Heterogeneous teams include UGVs and UAVs, have emerged as essential tools in planetary exploration due to their complementary capabilities [12]. While UGVs are well-suited for detailed surface exploration and full coverage of the terrain [13–15], UAVs provide an aerial perspective that enhances obstacle detection and expands environmental awareness [16]. Coordinating UAVs and UGVs for CPP, where the UGVs aim to fully explore a designated area while the UAV supports them by detecting obstacles, presents several key challenges, especially in an environment as unknown and dynamic as Mars [17].

CPP is a fundamental problem in robotics, where the goal is to explore an area efficiently while minimizing redundant coverage, overlaps and avoiding obstacles [18]. In Mars exploration, CPP becomes particularly complex due to the rugged, unpredictable terrain, the need for energy efficiency, and the critical importance of real-time obstacle detection and avoidance [19]. In this context, UGVs is supposed to complete full coverage of the area, while a UAV supports their operations by detecting obstacles, sharing environmental information, and assisting with task coordination [20]. However, achieving effective

coordination in such a multi-agent system introduces several challenges, particularly in terms of obstacle detection, energy management, and efficient task allocation [21].

One of the primary challenges is the unknown and dynamic nature of the Martian terrain [9]. The surface of Mars is filled with unpredictable features such as rocks, cliffs, and craters, which can impede UGV movement [22]. The UAV's aerial perspective allows it to detect these obstacles and provide real-time updates to the UGVs, improving their ability to navigate the terrain safely [23]. Another challenge is the UAV's limited battery life [24], it must periodically land on a UGV for recharging, and this requires careful coordination to ensure that recharging occurs efficiently without disrupting the UGVs' coverage tasks [25,26]. Additionally, effective multi-agent coordination is essential for ensuring that the UGVs cover the entire area without unnecessary overlap, while the UAVs assist in detecting obstacles and optimizing the UGVs' paths [27].

Historically, control systems for multi-agent coordination in robotics have often relied on centralized control architectures or predefined path planning algorithms [17,28]. Centralized systems, where a single controller directs all agents, can suffer from communication bottlenecks, particularly in large, rugged environments such as Mars, leading to delayed responses and degraded performance under dynamic conditions [29–31]. Predefined path planning methods, while effective in structured and well-mapped settings, are less suitable for unknown or evolving terrains, because they cannot readily adapt trajectories in real time when new obstacles or terrain features are encountered [32].

Alongside these centralized approaches, a substantial body of work addresses UAV–UGV coverage and energy logistics using classical search- and optimization-based formulations. Karapetyan et al. propose AG-CVG, which jointly plans UGV coverage and UAV tours with an on-board charger using mixed-integer optimization and heuristic refinement on a known grid map [25]. In AG-CVG, a single energy-constrained UAV and a single UGV first receive complete coverage paths that ignore energy constraints. These paths are then clustered and matched via mixed-integer programming and heuristics to schedule energy-feasible rendezvous, yielding reduced rendezvous overhead compared to a greedy baseline, but under fixed map and endurance assumptions. Lin et al. formulate persistent surveillance with energy-constrained UAVs and mobile charging stations as a robust optimization problem [33], partitioning the environment offline into maximal regions that each UAV can cover in one fuel cycle and designing cyclic UAV–UGV tours that minimize the worst-case inter-visit time while conservatively accounting for uncertain ground obstacles through a robustness parameter. Niu et al. cast UAV-assisted UGV path planning as a disciplined convex-concave program to coordinate aerial sensing with ground motion [34], and Wu et al. explore bio-inspired cooperative path planning for UAV–UGV teams in urban surveillance using metaheuristic search [35]. These approaches can produce high-quality solutions when the environment is well mapped and model parameters are reliable, but they remain largely offline, depend on fixed cost models, and require explicit re-planning triggers, which limits their adaptability under rapidly changing terrain, stochastic energy consumption, or perception errors typical of Mars-like settings.

Traditional heterogeneous robot planners normally rely on static, centrally generated paths that regard aerial and ground assets as loosely coupled subsystems. In the well-known look-ahead convoy strategy [36], used for bridge inspection and pipeline monitoring, a UAV proceeds along a predetermined racetrack ahead of a UGV, uploading waypoints to a ground-station optimizer that revises the UGV's lawn-mower trajectory only when a prescribed obstacle threshold is exceeded. Comparable leader–follower schemes are common in precision agriculture, where an autonomous tractor performs a deterministic boustrophedon sweep while a drone executes a scripted grid flight to gather multispectral imagery [37]. Although such plan-then-act paradigms succeed in slowly changing environments, they reveal three systemic weaknesses when faced with dynamic conditions. First, they are

combinatorially rigid, because integrating new UAV observations such as a freshly detected rockfall demands a large mixed-integer re-optimization that introduces delays unacceptable for time-critical missions [38]. Second, they impose heavy bandwidth requirements; continuous rebroadcast of amended ground paths or high-resolution aerial maps can saturate low-rate inter-robot links, especially in beyond-line-of-sight deployments [39]. Third, they disregard energy coupling, since fixed timetables rarely accommodate stochastic battery depletion; an unscheduled UAV diversion for recharging leaves the UGV convoy operating on outdated information, raising the likelihood of collisions and redundant coverage [40]. Lacking real-time task reallocation and recharge rescheduling, the performance of these frameworks degrades sharply in mission-critical scenarios such as Mars surface exploration, where terrain geometry and power budgets evolve unpredictably.

To address the limitations of traditional control methods in dynamic and uncertain environments like Mars, Reinforcement Learning (RL) presents an effective alternative for coordinating multiple agents [41]. RL allows agents to autonomously learn optimal policies through interaction with the environment, receiving feedback in the form of rewards and adjusting their actions accordingly [42]. RL has demonstrated success in various space applications [43], including satellite collision avoidance [44], satellite orbital maneuvering [45], satellite docking [46,47], satellite task scheduling [48,49], and satellite failure predictions [50]. RL has also shown strong potential in both single-agent and multi-agent path planning, particularly in unpredictable environments [51–53]. For Mars exploration, where UAVs and UGVs must collaborate to navigate unknown terrain, RL can optimize coordination, enabling agents to adapt to changes in terrain, manage energy constraints, and make real-time decisions related to obstacle avoidance and recharging [54].

Deep Q-Networks (DQN) [55] are suitable for discrete decision-making tasks, such as determining when and where the UAV should recharge. DQN enables the UAV to learn the optimal times to recharge based on its energy levels and proximity to UGVs, ensuring continuous operation without depleting energy [56]. This approach ensures UAV recharging is integrated smoothly into the exploration task, supporting uninterrupted UGV coverage.

When multiple agents collaborate, Multi-Agent Reinforcement Learning (MARL) [57] become particularly powerful. Compared to traditional control methods, RL provides real-time adaptability, allowing agents to dynamically adjust their strategies based on their current state and the environment. This makes RL particularly well-suited for complex missions in unstructured environments like Mars [58]. Research in multi-agent RL has demonstrated its effectiveness in improving autonomous exploration, task allocation, and energy efficiency [59]. Studies show that RL enables agents to learn cooperative behaviors that optimize coverage and avoid redundant actions, making it a strong candidate for managing UAV-UGV systems in planetary exploration missions [12,17,41].

Several RL studies have already explored heterogeneous robot teams. In one study authors introduced MADDPG with centralized critics and decentralized actors and demonstrated cooperative behavior among simulated agents with dissimilar dynamics, yet their method required continuous high-bandwidth exchange of global states during training and still relied on sparse team rewards that slowed convergence in complex domains [60]. Rashid et al. proposed QMIX to factorize a global value function into per-agent terms and reported improved scalability [61], although the nonlinear mixing network increased computation and offered only limited guidance for agents that possessed entirely different action repertoires such as aerial versus ground vehicles. More recent work by Qin et al. incorporated curriculum learning to ease heterogeneity but still assumed a shared battery model and therefore did not address asymmetric energy constraints in air-ground teams [62]. Sunehag et al. [63] then introduced Value Decomposition Networks (VDN), which decompose the team Q-function into a simple additive sum of individual value functions, enabling centralized training with

fully decentralized execution. VDN greatly reduces the communication and computational overhead compared to QMIX, and it provides a clear credit-assignment mechanism. However, it still depends on sparse global rewards and cannot by itself reconcile highly divergent local state or action spaces. These approaches mark important progress, yet several challenges persist.

Despite the progress achieved by these heterogeneous-team algorithms, key obstacles remain unresolved. Divergent state and action spaces continue to impede effective parameter sharing, causing instability when attempting to apply a single network architecture across ground and aerial agents. Non-stationarity grows more acute as each update from the UAV immediately shifts the value landscape perceived by the UGVs, slowing convergence and often leading to oscillatory behavior. Sparse, global team rewards obscure individual credit assignment, leaving agents uncertain about how their local decisions impact overall coverage performance. Traditional approaches that employ fully centralized critics demand continuous high-bandwidth exchanges of global observations and Q-values, a requirement that fails in bandwidth-constrained environments and scales poorly as team size increases [64]. Finally, most frameworks ignore the coupling between path planning and energy logistics, so an unscheduled drone landing for recharge can invalidate the ground vehicles' current plans and introduce coverage gaps [65].

To overcome these challenges, this study introduces a novel integration of VDN and individualized, team aligned potential-based reward shaping to explicitly guide heterogeneous decentralized agents towards globally optimal cooperative policies. The Centralized Feedback-Driven Reward Multi-Agent Reinforcement Learning (CFR-MARL) framework overcomes these persistent challenges in several ways. Each agent maintains its own DQN tailored to its unique state and action space, avoiding the pitfalls of monolithic architectures. A VDN aggregates local Q-values into a team-level potential that guides decentralized learning without continuous data exchange, and potential-based reward shaping furnishes each agent with dense, stepwise feedback that greatly improves credit assignment and accelerates convergence. Battery status is explicitly modelled in both the state representation and the reward function, ensuring that UAV recharge decisions trigger real-time replanning for UGVs rather than causing mission disruption. Parameter synchronization occurs only intermittently, reducing communication overhead while preserving sufficient global context for coordinated decision-making. As a result, the CFR-MARL achieves stable learning, dramatically faster coverage, minimal overlap and collision rates, and graceful scalability to larger heterogeneous teams operating under strict energy and bandwidth constraints.

Key contributions are:

- The CFR-MARL is applied to a Mars-like simulation featuring multiple UGVs and UAVs, using this heterogeneous multi-robot setup to validate the algorithm's effectiveness under coverage tasks.
- A Discretized Matrix is designed that fuses each agent's local observations into a unified grid representation.
- A novel framework is proposed that integrates centralized VDN team-value training with individualized, team-aligned potential-based reward shaping for heterogeneous agents, achieving accelerated convergence and robust decentralized coordination.
- By computing rewards from the difference in the global VDN potential, the CFR-MARL decouples continuous coordination from frequent network synchronization. Agents synchronize parameters only intermittently reducing run-time communication overhead.
- Simulation results demonstrate that CFR-MARL scales efficiently to larger heterogeneous teams, yielding faster coverage, lower overlap, and reduced collision rates under strict energy and bandwidth constraints.

By integrating these elements, the proposed method provides a scalable, adaptive solution for coordinating multi-agent systems in

complex environments like Mars. This approach addresses the key challenges of full area coverage, obstacle detection, and energy management, offering a robust framework for future planetary exploration missions.

The rest of the paper is structured as follows. Section 2 presents the overall system framework and problem definition for cooperative coverage path planning, including the discretized communication matrix (DM), and the formal problem definition with coverage and overlap metrics. Section 3 describes the proposed CFR-MARL method in detail. Section 4 describes the simulation setup and implementation details, and report training and testing results. Finally, Section 5 concludes the paper with a summary of contributions and potential directions for future work.

2. System framework and problem definition

In this section, a new system framework is proposed for CPP using a heterogeneous team of UGVs and UAVs, with the additional complexity of UAV recharging and decision-making for UGV assistance. Consider a heterogeneous team comprising N_G UGVs and N_A UAVs, with $N = N_G + N_A$ agents in total. UGVs are indexed by $i \in \{1, \dots, N_G\}$, and UAVs are indexed by $l \in \{1, \dots, N_A\}$. The system is designed to fully leverage the capabilities of the UAV for obstacle detection, coverage validation, and recharging, while the UGVs are responsible for complete area coverage. The UAV, with its aerial view, assists the UGVs by validating multiple regions of the environment at once, deciding when and where to recharge, and choosing which UGV to follow for recharging or assistance based on their visibility of the environment. Fig. 1 illustrates the environment and the picture of the problem.

2.1. UAV and UGV roles in the system

In this system, the UGVs are responsible for full exploration and

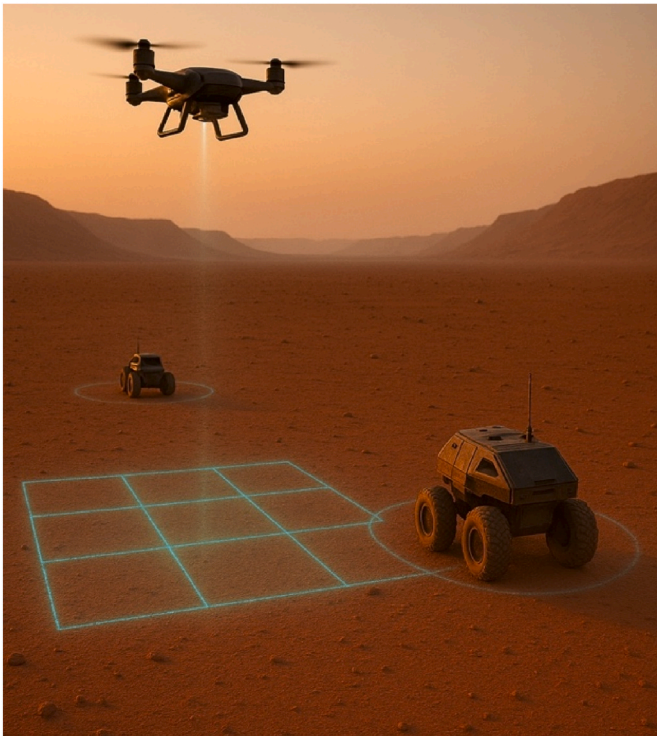


Fig. 1. System overview of the heterogeneous UAV-UGV coverage framework. A UAV continuously scans a local 3×3 grid sector and updates the shared DM, while UGVs traverse the Martian surface to achieve full coverage under CFR-MARL guidance.

coverage of the target area, moving through the terrain and ensuring that each region is explored. The UAV, on the other hand, has multiple roles:

- **Obstacle detection:** The UAV uses its aerial view to detect and validate obstacles that might obstruct the UGV's path.
- **Coverage validation:** The UAV can view a wider area (9 times more than the UGV) and thus assist in verifying that the UGV has covered all parts of the area.
- **Recharging:** The UAV needs to periodically land on a UGVs for recharging. This requires careful decision-making to select the most appropriate UGV for recharging based on proximity and the UGV's movement.
- **UGV assistance:** The UAV also assists UGVs by choosing which UGV to follow, dynamically providing updated obstacle information to the UGV to ensure efficient coverage.

2.2. Coverage with differentiated views

The environment is modelled as a $m \times m$ rectangle, with UAV and UGVs moving in discrete space. However, the system uses a discretized matrix (DM) to represent the environment and share information. Both UAVs and UGVs update this matrix in real time using their own local observations, but their views of the environment differ based on their altitude and sensing capabilities.

Each UGV can only observe the one grid cell directly around its current location. This limits the UGV's ability to detect distant obstacles or unexplored areas and makes it reliant on the UAV for more comprehensive obstacle detection and exploration data. The UAV has a field of view (FOV) that covers nine grid cells at a time (a 3×3 grid around its current location). This allows the UAV to simultaneously observe and validate multiple regions of the environment. UAV and UGVs can mark grid cells as covered, unexplored, or containing obstacles, providing real-time updates to each other.

2.3. Discretized UAV-UGV internal communication matrix

Environmental data is shared through a DM at discrete time t . This matrix represents the environment as a set of grid cells, where each cell holds information about the current state of the environment. At the beginning of each mission, the global DM is initialized as $DM(t_0) = -1$ for all cells, the terrain is treated as completely unknown. All information about free space, coverage, and obstacles is therefore acquired online from the agents' onboard sensing and fused into DM over time. In this study, the sensing processes are idealized. When a UAV or UGV observes a cell within its sensing footprint, the corresponding DM entry is assumed to reflect the true underlying terrain state (free or obstacle). Perception misclassifications are not explicitly modelled in the present simulations.

$$DM(t) \in \{-1, 0, 1, 3\}^{m \times m} \quad (1)$$

Each entry $DM_{u,v}(t)$ denotes the state of cell (u, v) at time t :

- Unexplored region: Value -1 .
- Explored by UAV: Value 0 .
- Explored by UGV: Value 1 .
- Obstacle detected: Value 3 .

At each timestep, each UAV_l and each UGV_i generate local observation matrices $DM^{(UAV_l)}(t)$ and $DM^{(UGV_i)}(t)$, respectively, where $i \in \{1, \dots, N_G\}$, and $l \in \{1, \dots, N_A\}$. Each observation matrix shares the same dimensions and value conventions as $DM(t)$, with observed states recorded in the relevant cells based on the sensing footprints. Cells outside each agent's sensing footprint are assigned the default "unexplored" state (-1).

All agents share the global $DM(t)$, which integrates these local observations. At each time step, the global DM is updated by taking the element-wise maximum among the existing global state and all local observations from the UAVs and UGVs. Formally, the updating rule for each cell (u, v) is:

$$DM_{u,v}(t+1) = \max \left\{ DM_{u,v}(t), \max_{i=1, \dots, N_G} DM_{u,v}^{UGV_i}(t+1), \max_{l=1, \dots, N_A} DM_{u,v}^{(UAV_l)}(t+1) \right\}. \quad (2)$$

where the element-wise maximum is performed over the ordered set of states $\{-1, 0, 1, 3\}$. Thus, once a higher-priority state is detected by any agent, it persists and cannot be overwritten by a lower-priority state. Fig. 2 illustrates the DM and how the system works.

2.4. Problem definition

The goal of this system is for UGVs to achieve full coverage of the target area, with UAV assisting by detecting obstacles. The total task completion time is determined by the UGV that takes the longest to fully explore its designated area. Each UGV starts from a different position and moves through the area, adjusting its path based on UAV-provided data.

2.4.1. Environment discretization and cell definition

To enable efficient path planning and coverage accounting, the continuous operational environment $\Omega \subset \mathbb{R}^2$ is discretized into a regular grid of cells. Let the environment be a square of side length L , which is partitioned into an $m \times m$ array of square cells of equal size. Although the current study considers a square environment discretized into an $m \times m$ grid, the same formulation applies directly to rectangular grids of size $m_x \times m_y$. The side length of each cell is thus given by $\Delta = \frac{L}{m}$.

Each cell in the grid is uniquely indexed by a pair of integers (u, v) , where $u, v \in \{1, 2, \dots, m\}$. The spatial domain of the (u, v) -th cell is defined as:

$$C_{u,v} = [(u-1)\Delta, u\Delta] \times [(v-1)\Delta, v\Delta] \quad (3)$$

Here $C_{u,v} \subset \mathbb{R}^2$ denotes the spatial domain (area) of cell (u, v) . such that the collection $\{C_{u,v}\}_{u,v=1}^m$ forms a partition of the environment Ω .

To associate a position $x = (x_1, x_2) \in \Omega$ with its corresponding grid cell, the cell-mapping function $cell: \Omega \rightarrow \{1, \dots, m\} \times \{1, \dots, m\}$ is defined as

$$cell(x) = (u, v) \text{ if and only if } x \in C_{u,v} \quad (4)$$

In closed form, this mapping is given by:

$$cell(x) = \left(\left\lceil \frac{x_1}{\Delta} \right\rceil, \left\lceil \frac{x_2}{\Delta} \right\rceil \right) \quad (5)$$

with the convention that $\left\lceil \frac{L}{\Delta} \right\rceil = m$.

The geometric centroid of cell (u, v) is denoted by $c_{u,v}$:

$$c_{u,v} = \left[\left(u - \frac{1}{2} \right) \Delta, \left(v - \frac{1}{2} \right) \Delta \right]. \quad (6)$$

At each time step t , the position of UGV_i , denoted as $x_i(t)$, is converted to its corresponding grid cell index via $cell(x_i(t))$. This mapping is essential for updating the DM and for evaluating area coverage and overlapping.

2.5. Coverage and overlap metrics

Each agent covers a set of grid cells over the mission duration. Let V_i denote the set of all grid cells it covers, and O_i the set of cells where its coverage overlaps with other same agent.

$$V_i = \{(u_i(k), v_i(k))\}_{k=0}^{K_i} \quad (7)$$

and let

$$O_i = V_i \cap \left(\bigcup_j V_j \right) \quad (8)$$

be its overlap with another same agent. The key metrics are defined as follows:

- Raw coverage area: $A_i^{cover} = |V_i| \cdot \Delta^2$
- Overlap area: $A_i^{overlap} = |O_i| \cdot \Delta^2$

These ensure that overlaps are penalized.

2.5.1. Time calculation for UAVs and UGVs

Each UGV_i traverses a sequence of cells to achieve full coverage of its assigned region. If it takes K_i moves to complete coverage, the total traversal time is:

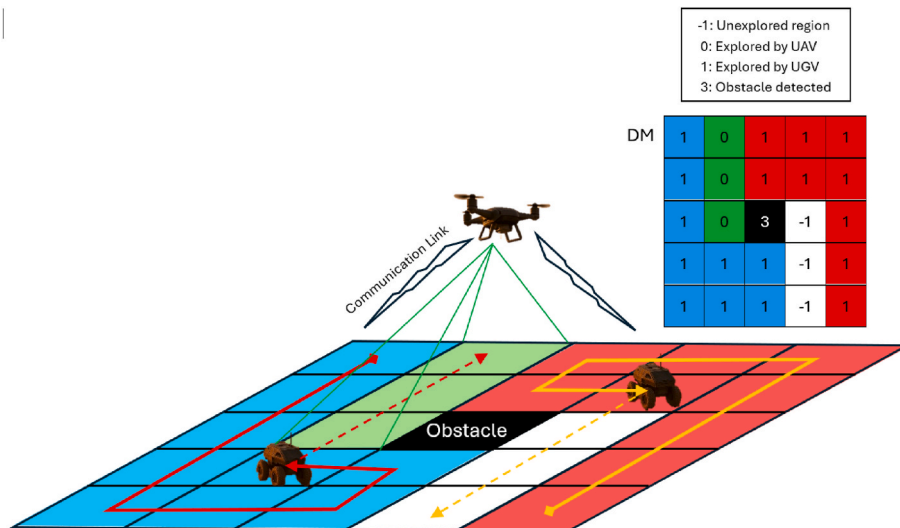


Fig. 2. Illustration of the DM update process. Each agent produces a local matrix, marking cells as unexplored, covered, or obstacle.

$$T_{UGV_i} = \sum_{k=1}^{K_i} \frac{\|c_{ni}(k) - c_{ni}(k-1)\|}{v_{UGV}} \quad (9)$$

where v_{UGV} is the UGV's speed and $c_i(k)$ is the centroid of the k -th cell visited.

The total coverage time for the mission contributed by the UGVs is the maximum time taken by any UGV to complete its task:

$$T_{completion}^{UGV} = \max_{i \in \{1, 2, \dots, N_G\}} T_{UGV_i} \quad (10)$$

If UAV_i visits a sequence $\{(u_i(k), v_i(k))\}_{k=0}^{K_i}$ and performs r_i recharges (each of duration T_{ch}), then

$$T_{UAV_i} = \sum_{k=1}^{K_i} \frac{\|c_{ni}(k) - c_{ni}(k-1)\|}{v_{UAV}} + r_i T_{ch} \quad (11)$$

The total coverage time is the maximum time of both.

2.5.2. UGV objective functions

The goal is for the UGVs, with UAV assistance, to collectively cover the entire area with minimal overlap. This ensures efficient and complete exploration of the target area while avoiding unnecessary duplication in coverage.

The UGV objective function is formulated as:

$$\min_{K_i}^{i=1, 2, \dots, N_G} \left(\max_{i=1, \dots, N_G} (T_{UGV_i}) + w \sum_{i=1}^{N_G} A_i^{overlap} \right), \quad (12)$$

where $w > 0$ denote a weighting factor balancing the trade-off between coverage efficiency and overlap minimization.

The objective function minimizes the mission makespan and penalizes overlap, subject to the constraint of full coverage of the free space. All UGVs must remain within the operational grid and are prohibited from entering obstacle cells. This formulation ensures that the UGV fleet achieves both rapid and efficient area coverage, while explicitly discouraging redundant traversal of the same grid cells.

2.5.3. UAV objective functions

The primary objective for the UAV is to achieve complete and efficient coverage of the target area in minimal time, while minimizing unnecessary overlap by itself and energy constraints. Unlike UGVs, UAV do not encounter static terrain obstacles within their flight domain.

For each UAV_i CPP optimization is:

$$\min (T_{UAV_i} + w A_{UAV_i}^{overlap}) \quad (13)$$

subject to:

- (i) Every free cell is visited at least once: $V_{UAV} \supseteq F$
- (ii) For each sequence between two consecutive recharges, the total energy consumed must not exceed the UAV's battery capacity E_{max}

Where $V_{UAV} = \bigcup_{i=1}^{N_A} V_{UAV_i}$ denotes the union of grid cells visited by all UAVs.

3. Proposed method

In this section, the CFR-MARL (Centralized Feedback-Driven Reward Multi-Agent Reinforcement Learning) framework is presented for decentralized cooperative CPP by heterogeneous agents on Mars. The proposed CFR-MARL framework enables fully decentralized execution of heterogeneous agents (UGVs and UAVs) for CPP on Mars, while leveraging centralized training via a team-level value function and reward shaping.

The core principle of the CFR-MARL is combining the flexibility and

robustness of decentralized policies which allow each heterogeneous agent to learn from its unique local observations and action spaces with the strategic advantage of centralized training. Specifically, the CFR-MARL incorporates centralized feedback through a Value Decomposition Network (VDN) to guide decentralized policies via potential-based reward shaping.

The primary benefits include improved coordination efficiency, faster convergence, scalability to larger teams of heterogeneous agents, and decentralized execution without sacrificing cooperative optimality.

Furthermore, to ensure effective coordination and synchronization among the agents in the proposed framework, a unified time step is implemented that all agents adhere to. At each discrete time step t , the UAV and the UGVs simultaneously observe their respective states, select and execute their actions, receive rewards, and update the DM. Control signals for the agents are updated every Δt seconds, where Δt is the time step. The schematic diagram of the CFR-MARL is illustrated in Fig. 3.

The following subsections first describe the state-action-reward design for each agent (Sec. 3.1), then detail the VDN-based centralized training (Sec. 3.2), introduce the potential-based shaping (Sec. 3.3), and finally introduce the CFR-MARL framework (Sec. 3.4).

3.1. Markov Decision Process configuration

To enable autonomous decision-making, the CPP problem for both UAV and UGVs is modelled as a Markov Decision Process (MDP) [66], allowing each agent to learn optimal policies through interaction with the environment.

3.1.1. MDP for UAV

The state of the UAV_i at time t is defined as:

$$S_{UAV_i}(t) = (P_{UAV_i}(t), \{P_{UGV_i}(t)\}_{i=1}^{N_G}, DM(t), E_i(t)), \quad (14)$$

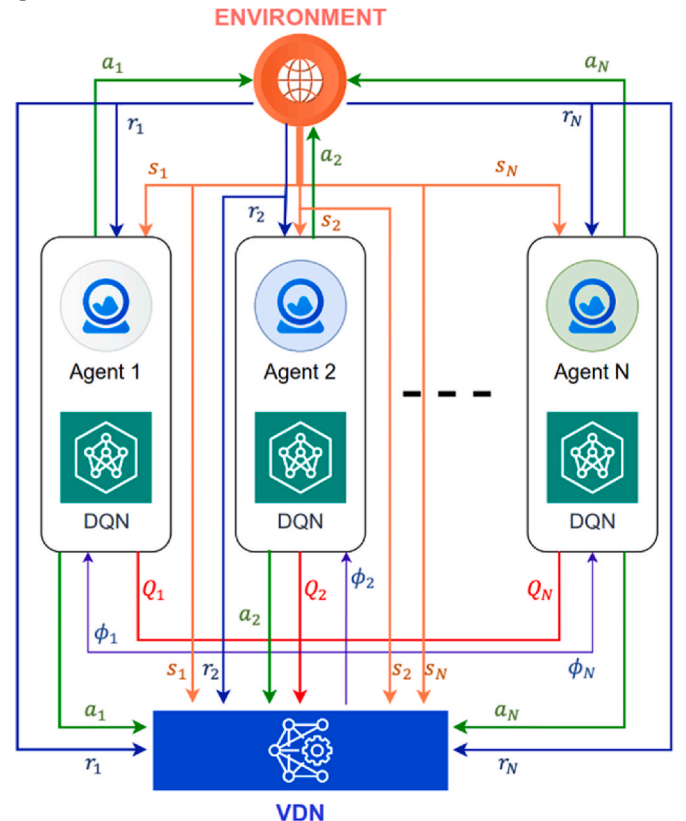


Fig. 3. Schematic of the CFR-MARL framework.

where, $E_i(t)$ represents the current energy level of the UAV_{*i*}, $P_{UAV_i}(t)$ is the UAV_{*i*}'s current position, and P_{UGV_i} are the positions of the all UGVs. $DM(t)$ refers to the last update of the DM matrix.

The UAV_{*i*}'s action space A_{UAV_i} consists of discrete actions, including movement between adjacent grid cells (up, down, left, right), assisting each UGV_{*i*}, and choosing to recharge on a UGV_{*i*}. State transitions depend on the UAV's movement, energy dynamics, and updates to the DM based on its observations. The UAV's action space A_{UAV} are defined as:

$$A_{UAV_i} : \{\uparrow, \downarrow, \leftarrow, \rightarrow, assist(UGV_i), recharge(UGV_i)\}$$

The UAV's reward function is designed to encourage effective assistance and energy management:

$$R_{UAV_i}(t) = w_{1u} \cdot r_{a,i}(t) + w_{2u} \cdot r_{c,i}(t) + w_{3u} \cdot r_{e,i}(t) \quad (15)$$

where the weighting factors w_{1u} , w_{2u} , w_{3u} balance the objectives of agent assistance, coverage efficiency, and energy management, respectively.

The term $r_a(t)$, rewards the UAV for assisting the UGV with the greatest immediate exploration need. At each timestep, the exploration needs of UGV_{*i*} is defined as the number of unexplored cells ($DM_{u,v} = -1$) within a radius R around its current position $P_{UGV_i}(t)$:

$$U_i(t) = \{(u, v) \mid |u - p_u^{UGV_i}(t)| + |v - p_v^{UGV_i}(t)| \leq R, DM_{u,v} = -1\} \quad (16)$$

If the UAV chooses action $assist(UGV_i)$, (fly to support UGV_{*i*}), the assistance reward is given by:

$$r_{a,i}(t) = \frac{U_i(t)}{\max_j U_j(t) + \epsilon}, (\epsilon > 0) \quad (17)$$

This formulation encourages the UAV to prioritize assisting the agent with the greatest number of unexplored neighboring cells. The coverage reward $r_c(t)$, awards the UAV for exploring previously unexplored grid cells:

$$r_{c,i}(t) = \begin{cases} +1 & \text{if UAV explores a new cell} \\ 0 & \text{otherwise.} \end{cases} \quad (18)$$

Efficient energy management is crucial given the UAV's limited battery capacity. The energy reward $r_e(t)$ is defined as:

$$r_{e,i}(t) = \begin{cases} +1 & \text{if UAV recharges timely} \\ -10 & \text{if battery fully depleted without recharging,} \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

This ensures that the UAV proactively manages its battery state to maintain continuous operation without interruptions due to energy constraints.

3.1.2. MDP for UGVs

The UGVs decision-making is based on their position, velocity, and information from the DM. The state space for UGV_{*i*}, at time t is represented as:

$$S_{UGV_i}(t) = (P_{UGV_i}(t), DM(t)) \quad (20)$$

where $P_{UGV_i}(t)$, is the current position of UGV_{*i*}, and DM represents the last update of the shared communication matrix, which informs the UGVs about covered and unexplored regions.

The action space A_{UGV} includes:

$$A_{UGV_i} : \{\uparrow, \downarrow, \leftarrow, \rightarrow, \}$$

The UGV's reward function is designed to prioritize efficient coverage:

$$R_{UGV_i}(t) = w_{1c} \cdot R_c(t) + w_{2c} \cdot R_o(t) \quad (22)$$

where $R_c(t) = +1$ is the reward for covering new areas, and $R_o(t) = -1$ is a penalty for revisiting already covered areas and $w_1 > 0$, $w_2 > 0$. The

UGVs adjust their paths to maximize new coverage while minimizing overlap, ensuring full exploration of the target area.

3.2. Multi agent deep Q-network (MADQN)

The agents operate in a discrete action space, making the MADQN algorithm an appropriate choice for learning optimal policies. MADQN enables the agents to make decisions by approximating the optimal action-value function $Q(s, a)$ using a deep neural network. Each agent i is equipped with its own DQN. MADQN thus consists of N independent DQNs learning in parallel. For notational simplicity, in the following subsections all agents (UGVs and UAVs) are indexed by $i \in \{1, \dots, N\}$, where $N = N_G + N_A$.

Each agent maintains a DQN parameterized by θ_i , outputting Q-values for all actions, $Q_i(s_i, a_i; \theta_i)$. Each agent stores experience tuples $(s_i(t), a_i(t), r_i'(t), s_i(t+1))$ in a replay buffer D_i , where $r_i'(t)$ is the shaped reward (section 3.3). Parameters θ_i are updated by minimizing the temporal-difference loss:

$$L(\theta_i) = \mathbb{E}_{(s(t), a(t), r'(t), s(t+1)) \sim D_i} [(y_i - Q_i(s, a; \theta_i))^2] \quad (23)$$

where the TD-target y_i is computed with a target network θ_i^- , where denotes the target network:

$$y_i = r_i'(t) + \gamma Q_i(s_i(t+1), \arg\max_a Q_i(s_i(t+1), a; \theta_i); \theta_i^-) \quad (24)$$

At each time step t , the agents use the current state as input to the neural network, and the DQN outputs the Q-values for all possible actions. The loss function is computed as the difference between the target Q-value y_i and the predicted Q-value from the current network. Every C gradient steps, synchronize the target network:

$$\theta_i^- \leftarrow \theta_i \quad (25)$$

At runtime, agent i selects actions via an ϵ -greedy policy on its own Q-network.

In Table 1, a detailed summary is provided of the hyperparameters and network architecture used in the MADQN.

3.3. Centralized Value Decomposition Network (VDN)

To promote effective cooperation among heterogeneous agents, a Centralized Value Decomposition Network is employed as the backbone of the multi-agent training framework. The key motivation for choosing VDN is its ability to provide a scalable, interpretable, and communication-efficient structure for multi-agent reinforcement learning, particularly in scenarios where decentralized execution is essential but centralized training is feasible.

The core idea of VDN is to decompose the global team action-value function, Q_{tot} , to the sum of individual agent-specific Q_i , each of which is conditioned only on the local observation and action of that agent. Specifically, for a team of N agents, the joint action-value is written as:

Table 1
The list of the Hyperparameter.

Hyperparameter	Value
Learning Rate	0.001
Discount Factor	0.99
Exploration Strategy	ϵ -greedy policy
Initial ϵ	1.0
Final ϵ	0.01
Decay Rate	Linear decay over 10,000 steps
Replay Buffer Size	50,000 transitions
Mini-Batch Size	64
Target Network Update Frequency	Every 1000 steps
Loss Function	Mean Squared Error
Optimizer	Adam

$$Q_{\text{tot}}(S, A) = \sum_{i=1}^N Q_i(s_i, a_i; \theta_i) \quad (26)$$

where, S denotes the joint state, and A denotes the joint action. During training, experience tuples of the form $(s(t), a(t), r_{\text{team}}(t), s(t+1))$ are collected and stored in a centralized replay buffer D_{team} that aggregates the full joint transitions in individual replay buffer D_i , where the team reward is computed as the sum of intrinsic rewards from all agents:

$$r_{\text{team}}(t) = \sum_{i=1}^N r_i \quad (27)$$

The temporal difference (TD) target for centralized learning is given by:

$$y_i = r_{\text{team}}(t) + \gamma \max_{a'} Q_{\text{tot}}(S(t+1), A') \quad (28)$$

where γ is the discount factor, and the maximization is over all possible joint actions A' at the next state $S(t+1)$. The loss is minimized with respect to the parameters of each agent's Q-network, with gradients back-propagated through the additive structure.

By leveraging the VDN approach, the proposed framework achieves effective cooperation and sample-efficient learning, while maintaining the simplicity and robustness necessary for large-scale, real-world deployment of heterogeneous multi-agent systems.

While more powerful value-factorization networks such as QMIX [17] can capture non-linear interactions and provide richer global information to each agent during synchronization, they introduce an additional mixing network and incur significantly higher computational and communication overhead during both training and execution. In contrast, VDN's purely additive structure requires only a simple summation of individual Q-values, leading to much faster forward/backward passes and lower memory use. Although QMIX's mixing network can enhance each agent's awareness of the global team value, in resource-constrained planetary exploration scenarios the added synchronization and computational costs can outweigh these benefits, making VDN the preferred choice for efficient and scalable multi-agent coordination.

3.4. Centralized feedback-driven reward multi-agent reinforcement learning (CFR-MARL)

In this subsection, the proposed CFR-MARL method is formally introduced. The innovation of CFR-MARL is integrating a centralized team-level value function, derived via a VDN, with individual potential-based reward shaping to guide decentralized learning among heterogeneous agents. The theoretical foundation, the definition, and the mathematical formulation are detailed below.

In MARL, achieving coordinated behavior in decentralized execution scenarios remains challenging, particularly when the agents are heterogeneous in terms of state observations, action spaces, and reward structures. Standard decentralized methods often struggle due to sparse rewards and limited local observations, leading to slow convergence and suboptimal cooperation.

To address this, the CFR-MARL explicitly utilizes the centralized team-level value, defined via the VDN structure, as an individualized shaping potential function for reward shaping. Crucially, potential-based reward shaping guarantees that the optimal policy remains unchanged, provided the potential depends only on the state. This property enables us to introduce dense, global feedback signals without altering the theoretical optimality of policies.

A global potential function, $\Phi_{\text{global}}(S(t))$ is defined, based explicitly on the VDN formulation:

$$\Phi_{\text{global}}(S(t)) = \sum_{i=1}^N \max_{a_i} Q_i(s_i(t), a_i; \theta_i) \quad (29)$$

where $Q_i(s_i(t), a_i; \theta_i)$ is the state-action value function for agent i , parameterized by neural network parameters θ_i . This potential function represents the maximum possible team-level value achievable from state $S(t)$, assuming each agent individually selects its locally optimal action.

Each agent's local potential is similarly defined as:

$$\Phi_i(s_i(t)) = \max_{a_i} Q_i(s_i(t), a_i; \theta_i), i \in \{1, \dots, N\}. \quad (30)$$

This depends only on agent i 's local state s_i (not on the chosen action), thus meeting the state-only requirement for potential-based reward shaping. Each agent's individualized shaping potential Ψ_i is defined as follows:

$$\Psi_i(S(t)) = \Phi_{\text{global}}(S(t)) - \Phi_i(s_i(t)). \quad (31)$$

Moreover, since all Q_i are bounded, there exist finite constants $Q_{\min}$, $Q_{\max}$ such that

$$Q_{\min} \leq Q_i(s_i, a_i) \leq Q_{\max} \quad (32)$$

for all s_i and a_i . Therefore, the global potential is bounded:

$$(N-1)Q_{\min} \leq \Psi_i(S(t)) \leq (N-1)Q_{\max} \quad (33)$$

Since Ψ_i depends only on the joint state S (and not on the chosen action), it satisfies the state-only requirement for potential-based shaping. The shaping term is defined as:

$$F_i(t) = \gamma \Psi_i(S(t+1)) - \Psi_i(S(t)) \quad (34)$$

when added to each agent's reward, preserves the set of optimal policies. The formal proof of this invariance is provided in Ref. [67].

Given this potential function, the shaped reward $r'_i(t)$ provided to each agent i at time step t by adding a potential-based reward shaping term to its intrinsic reward $r_i^{\text{int}}(t)$ as:

$$r'_i(t) = r_i^{\text{int}}(t) + \alpha F_i(t) \quad (35)$$

where $r_i^{\text{int}}(t)$ is the intrinsic local reward specific to agent i 's task, and $\alpha > 0$ is a scaling hyperparameter that controls the influence of the potential-based shaping term.

Lemma 1. (Policy Invariance under Potential-Based Reward Shaping)

Let $M = (S, A, P, r, \gamma)$ be a MDP with state space S , action space A , transition probability P , reward function r , and discount factor $\gamma \in [0, 1]$. Let $\psi_i : S \rightarrow \mathbb{R}$ be a bounded potential function. Define the shaped reward as

$$r'(s, a, s') = r(s, a, s') + \gamma \psi_i(s') - \psi_i(s) \quad (36)$$

Then, the set of optimal policies for the shaped reward r' is identical to the set of optimal policies for the original reward r .

Proof.

Let Q^* denote the optimal action-value function for r , satisfying the Bellman optimality equation:

$$Q^*(s, a) = \mathbb{E}_{s'} \left[r(s, a, s') + \gamma \max_{a'} Q^*(s', a') \right] \quad (37)$$

Let Q^* denote the optimal Q-function for the shaped reward r' :

$$Q^*(s, a) = \mathbb{E}_{s'} \left[r'(s, a, s') + \gamma \max_{a'} Q^*(s', a') \right] \quad (38)$$

Substituting for $r'(s, a, s')$, yields

$$Q^*(s, a) = \mathbb{E}_{s'} \left[r(s, a, s') + \gamma \psi_i(s') - \psi_i(s) + \gamma \max_{a'} Q^*(s', a') \right] \quad (39)$$

Let $\tilde{Q}(s, a) = Q^*(s, a) + \psi_i(s)$. Then,

$$\tilde{Q}(s, a) = \mathbb{E}_s \left[r(s, a, s') + \gamma \max_{a'} \tilde{Q}(s', a') \right] \quad (40)$$

which is the Bellman optimality equation for the original reward r . Therefore, $\tilde{Q}(s, a) = Q^*(s, a)$ and thus $Q^*(s, a) = Q^*(s, a) - \psi_i(s)$.

Since $\psi_i(s)$ is independent of a , it does not affect the maximization over a . Therefore, the set of optimal policies remains unchanged.

End.

This lemma guarantees that the CFR-MARL shaping method does not compromise the optimality of learned policies, thus ensuring theoretically coordination among agents.

Lemma 2. (Stability and Convergence of Q-Learning with Potential-Based Shaping)

Assume standard Robbins–Monro learning rate conditions and a finite MDP. Then, Q-learning with shaped reward

$$r'(s, a, s') = r(s, a, s') + \gamma \psi_i(s') - \psi_i(s) \quad (41)$$

converges almost surely to the optimal Q-function $Q^*(s, a)$. Furthermore, the optimal Q-function for the original reward can be recovered as $Q^*(s, a) = Q^*(s, a) + \psi_i(s)$.

The shaping term provides immediate feedback to each agent, quantifying precisely how their collective action at each step improves or degrades the global VDN-based potential function. Consequently, each agent learns to implicitly account for its teammates' potential contributions to the overall mission, driving faster convergence to effective coordinated behavior.

The CFR-MARL, through explicit centralized feedback via potential-based reward shaping, effectively bridges the gap between decentralized learning and centralized cooperative objectives. This formulation ensures optimality, accelerates convergence, and facilitates scalable and robust coordination among heterogeneous agents, thereby significantly advancing multi-agent reinforcement learning methodologies for complex exploration tasks such as Mars surface coverage.

In practice, the agents' DQNs are trained offline in simulation, but the resulting policies are executed fully online, at each discrete time step, every UAV and UGV selects its action based only on its current local state and the latest DM, without solving any mixed-integer optimization or replanning global tours. This contrasts with plan-then-execute frameworks that recompute joint coverage and rendezvous schedules whenever the environment or energy model changes.

Corollary 1 (Greedy Alignment of Shaped Q-Functions)

Let

$$Q'_i(s_i, a_i) = Q_i(s_i, a_i) + \alpha[\gamma \Psi_i(S') - \Psi_i(S)], \quad (42)$$

where Ψ_i is the individualized shaping potential. Then for any joint state $S = (s_1, \dots, s_N)$,

$$\sum_{i=1}^N \max_{a_i} Q'_i(s_i, a_i) = \max_{A=(a_1, \dots, a_N)} \sum_{i=1}^N Q_i(s_i, a_i) + \text{const} \quad (43)$$

up to an additive constant that does not depend on the chosen actions. Hence, if each agent picks

$a_i^* = \arg\max_{a_i} Q'_i(s_i, a_i)$, the resulting joint action maximizes the global

team value $\sum_{i=1}^N Q_i(s_i, a_i)$.

Proof.

Because

$$\psi_i(S) = \sum_{k \neq i} Q_k(s_k, a_k) - \max_{a_i} Q_i(s_i, a_i). \quad (44)$$

When one sums $\max_{a_i} Q'_i$ across agents, the ψ_i terms rearrange into a constant offset plus $\sum_i \max_{a_i} Q_i$. Those offsets cancel in any argmax, so greedy action selection under Q'_i aligns with maximizing the global Q-

sum.

Remark 1 (Shaping Computational Cost)

Each agent i , computes a local $\max_{a_i} Q_i(s_i, a_i)$ over its action set A_i , then sums these values across all agents. This incurs $O(\sum_i |A_i|)$ computation per time step, ensuring that CFR-MARL remains efficient and scalable as the team size grows.

Remark 2 (Selection of VDN Synchronization Frequency)

In the proposed CFR-MARL framework, the centralized VDN parameters are synchronized at a fixed frequency of every $C = 50$ time steps.

3.5. Hyperparameter and network architecture

For each agent, the DQN architecture consists of 3 fully connected layers. The first layer, or Layer 1, contains 256 neurons and uses the ReLU activation function to introduce non-linearity into the network, enabling it to capture complex patterns in the data. Following this, Layer 2 includes 128 neurons, also utilizing ReLU activation, further refining the network's ability to model the environment's dynamics. The third layer, Layer 3, consists of 64 neurons, continuing with the ReLU activation to condense the representation of the input state into more meaningful features. Finally, the Output Layer produces the Q-values, with each output neuron corresponding to a possible action the agent can take, representing the estimated value of each action in the current state.

4. Simulation study and results

This section provides an evaluation of the proposed framework, designed for coordinating two UGVs and one UAV for continuous CPP in a Mars exploration context. The simulations assess the system's performance in terms of coverage efficiency, energy management, collision avoidance, and multi-agent coordination under simulated Martian terrain conditions.

4.1. Simulation setup

The simulation models a $180 \text{ m} \times 180 \text{ m}$ area, discretized into a 36×36 grid to form the Discretized Communication Data Matrix, where each cell represents a $5 \text{ m} \times 5 \text{ m}$ region. This grid structure is designed to align with the UAV's field of view, facilitating efficient integration of the UAV's observations. Obstacles, which occupy 10% of the grid, simulate Mars's rocky and uneven terrain. The exact locations of these obstacles are not revealed to the agents. At the start of each episode, the shared DM is fully unexplored, and both UAVs and UGVs must discover free cells and obstacles solely through their local sensing footprints. Thus, all coverage decisions are taken in an initially unknown environment. In these simulations, the UAV camera and the UGV local obstacle sensors are modelled as perfect occupancy detectors for cells within their footprints. When a cell is observed, its DM label is assumed to match the true free or obstacle state, and misclassifications are not explicitly simulated. Communication is assumed to be ideal, to focus on the agents' coordination and decision-making capabilities without considering delays or interruptions.

The UAV operates at a fixed altitude and moves between grid cells at a constant speed of 5 m/s, capable of moving in four directions. The UAV can observe a 3×3 grid area ($15 \text{ m} \times 15 \text{ m}$) at its current position, allowing it to detect obstacles and validate coverage over a broader region. The UAV has a battery life allowing for 25 min of continuous flight and consumes energy proportionally to its movement and observation activities. It can land on a UGV for a 5-min recharge. In the simulations, the recharging process (including the transit towards the UGV, landing, and the recharging time) is modelled separately by the MATLAB environment and treated as part of the environment dynamics. The proposed CFR-MARL controller only decides when recharging is

needed and which UGV to use. When the recharge on UGV action is selected and the UAV already occupies the same grid cell as that UGV, the UAV is assumed to land, remain docked for the fixed recharge duration of 5 min, and have its battery reset to full, while the UGV continues moving. During this docked phase, the UAV remains stationary and is unavailable for sensing or assistance actions. The UAV is equipped with a camera for terrain observation, coverage validation, and obstacle detection.

UGVs use a differential drive system, which enables precise control over steering and acceleration. They operate in continuous space and navigate at a maximum speed of 1 m/s. UGVs are equipped with Inertial Measurement Units (IMUs) for accurate positioning. They rely on local sensing for observing nearby grid cells and on the UAV for detecting obstacles beyond their immediate range.

4.2. Implementation details

All simulations were conducted using MATLAB R2023b, utilizing the Reinforcement Learning Toolbox for agent training, the Deep Learning Toolbox for neural network construction and optimization, and the Robotics System Toolbox for agent dynamics and environmental modelling. Each agent was modelled as an independent DQN, as described in Section 3.4, with three fully connected layers (256, 128, 64 neurons respectively, ReLU activations) and a linear output layer representing all available actions. The Adam optimizer was used with a learning rate of 0.001, discount factor $\gamma = 0.99$, replay buffer size of 50,000 transitions, and minibatch size of 64. Target networks were updated every 1000 gradient steps. Agents employed an ϵ -greedy exploration schedule, decaying linearly from $\epsilon = 1.0$ to $\epsilon = 0.01$ over the first 10,000 steps. All weighting factors are treated as design hyperparameters. Their values are selected empirically from preliminary simulations on separate randomly generated maps to obtain consistent behavior and a reasonable trade-off between coverage efficiency, overlap and energy usage, following standard practice in RL literature.

Agents were trained for 12,000 episodes, with each episode terminating upon full coverage or after a maximum of 4500 timesteps ($\Delta t = 1$ s). Start positions of all agents and obstacle locations were randomized uniformly in each episode to promote robustness and generalization. All simulations were conducted over five independent random seeds, and results are reported as mean.

4.3. Evaluation metrics

To assess the framework's performance, the following metrics are used. Total Coverage Time (TC), defined as the time taken by the agents to cover the entire designated area. Overlap Ratio (OR), the ratio of overlapped areas to the total coverage area, indicating redundant coverage. Success Rate (SR), the percentage of episodes in which the mission was completed without collisions or UAV battery depletion. Collision Rate (CR), the frequency of collisions between UGVs or with obstacles, and Average Reward per Episode (AR), which tracks the performance of the agents during training. All reported metrics represent the mean value, averaged over five independent random seeds.

For comparative evaluation, the CFR-MARL framework is benchmarked against two baseline methods, both employing identical neural network architectures and training hyperparameters. The first baseline, MADQN, involves each agent learning a policy via its own DQN using only intrinsic local rewards, with the DM for information sharing. The second baseline, Centralized VDN, implements a single VDN critic over the full joint state and action space, trained with team rewards as previously defined, also utilizing the DM. Additionally, the proposed method is compared against a baseline using only UGVs without UAV support, to further evaluate the contribution of heterogeneous agent cooperation and is finally compared with the QMIX with utilizing DM.

4.4. Simulation results

4.4.1. Training performance and convergence analysis

To evaluate the convergence behavior and learning stability of the proposed CFR-MARL framework, simulations were conducted across 12,000 training episodes, averaging results from five independent random seeds. Fig. 4 shows the learning curves for the proposed CFR-MARL framework compared to the baseline methods, MADQN (with DM), VDN (with DM), QMIX (with DM) and the CFR-MARL (no UAV). Each data point represents the mean average reward per episode over intervals of 500 episodes, with shaded regions indicating one standard deviation to illustrate stability and variance clearly.

The proposed CFR-MARL method exhibits superior convergence performance, reaching a stable average reward of approximately 260. Among the baselines, QMIX converges next-best at around 228, followed by CFR-MARL (no UAV) at 208, VDN + DM at 176, and MADQN + DM at 125. For VDN + DM and MADQN + DM, these values correspond to their final plateau levels after extending training to approximately 15,000 episodes, at which point their learning curves no longer exhibit a consistent upward trend. This ranking demonstrates that while the nonlinear mixing in QMIX provides a boost over VDN, the potential-based shaping combined with centralized feedback yields the highest and most stable returns, particularly when leveraging UAV-UGV cooperation.

Periodic frozen-policy evaluations ($\epsilon = 0$) at 2000-episode intervals during training reveal a clear trend in the progression of success rates for all algorithms, as illustrated in Fig. 5. The results show that MADQN (with DM) exhibits the slowest improvement, while VDN (with DM) and QMIX (with DM) demonstrate moderate and faster increases in success rate, respectively. The CFR-MARL (no UAV) shows a rapid rise in reliability due to dense potential-based reward shaping, and the full the CFR-MARL framework with benefiting from both shaping and UAV-UGV cooperation, achieves the fastest and most robust learning.

4.4.2. Testing results

The learned policies were further evaluated by running simulation tests under the same measurement conditions, with increasing the obstacle density up to 15% to measure the efficacy, reliability, and robustness of the proposed framework. Table 2 summarizes the key performance metrics, averaged across five independent runs, including mean values and their respective standard deviations.

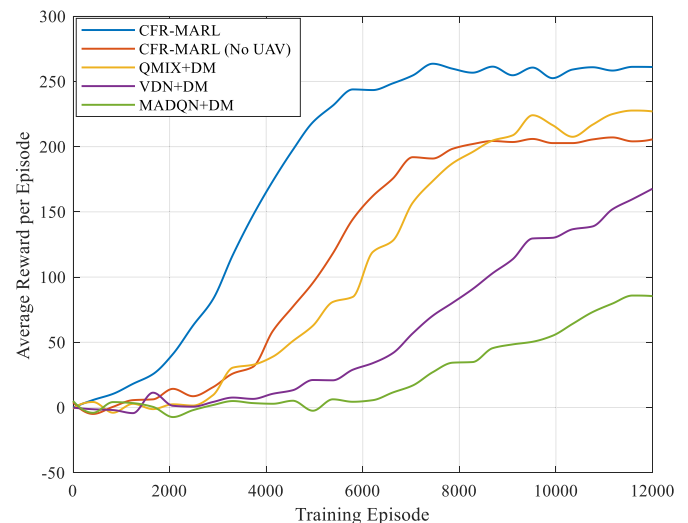


Fig. 4. The learning curves for the proposed CFR-MARL framework compared to the baseline methods.

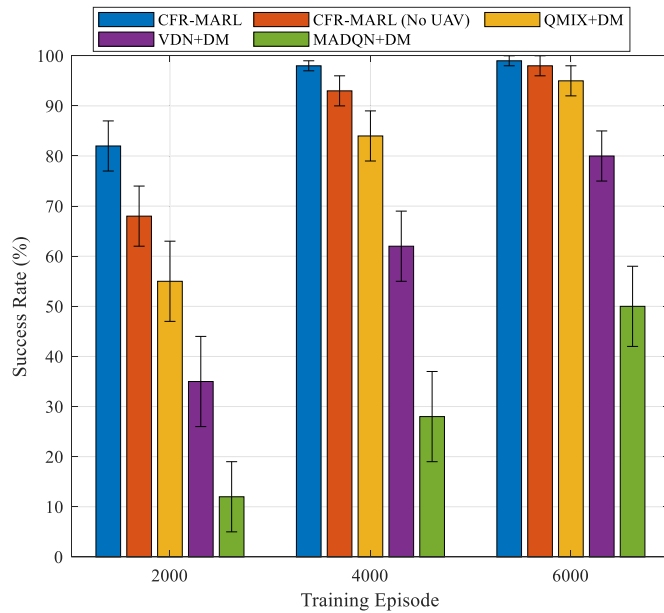


Fig. 5. Periodic success rate during training (10% obstacles).

Table 2
Performance metrics comparison.

Method	TC (s)	OR (%)	SR (%)	CR (%)	AR
MADQN	4069.5 ± 140.0	18.81 ± 0.66	77.11 ± 4.64	13.59 ± 0.45	120.04 ± 7.66
VDN	3669.7 ± 129.5	14.87 ± 0.49	88.42 ± 2.67	8.66 ± 0.31	168.64 ± 5.75
QMIX	3443.2 ± 122.5	7.05 ± 0.25	91.78 ± 3.24	2.89 ± 0.11	211.51 ± 7.15
CFR-MARL	3574.2 ± 119.0	9.74 ± 0.35	93.39 ± 2.74	5.42 ± 0.20	198.18 ± 3.83
CFR-MARL (no UAV)	3278.6 ± 112.0	4.86 ± 0.17	98.53 ± 2.45	0.92 ± 0.07	246.51 ± 8.80

4.5. Results analysis and discussion

The proposed CFR-MARL framework reduces the total coverage time, outperforming MADQN by roughly 20%. The improvement stems from centralized feedback-driven reward shaping, resulting in more coordinated and strategic exploration.

The CFR-MARL achieves a notably low overlap ratio, reducing redundancy by approximately 74% compared to MADQN. This highlights the effectiveness of UAV integration in optimizing exploration tasks, ensuring minimal redundant coverage.

The success rate under CFR-MARL is exceptionally high, demonstrating superior reliability in achieving mission goals without collisions or energy depletion. Correspondingly, the collision rate decreases dramatically, emphasizing the UAV's critical role in obstacle detection and avoidance strategies.

The CFR-MARL method yields the highest average reward per episode higher than the next-best variant (CFR-MARL without UAV). This confirms that integrating UAV assistance and centralized potential-based reward shaping enhances individual agent performance and overall team effectiveness.

To quantify the contribution of the UAV explicitly, The CFR-MARL is compared with its variant without UAV assistance (CFR-MARL no UAV). The results clearly illustrate the UAV's positive impact on all metrics. This ablation confirms the critical strategic value of UAV-UGV heterogeneity and collaboration for Mars surface exploration missions. Moreover, compared to a nonlinear mixer such as QMIX, the CFR-MARL completes coverage about 7% faster and attains about 17% higher

average reward, demonstrating that potential-based shaping adds meaningful gains over advanced value-factorization alone.

Statistically improvements across multiple metrics demonstrate the robustness and scalability of the CFR-MARL framework. Practically, these results indicate substantial potential to improve real-world mission efficiency, reliability, and safety, reducing resource consumption and mission risks. By leveraging decentralized execution with centralized training and feedback mechanisms, CFR-MARL strikes an optimal balance between autonomy and coordinated strategy, making it highly suitable for deployment in realistic planetary exploration scenarios.

Compared to classical UAV–UGV teaming methods such as AG-CVG [25] and the robust planning framework of Lin et al. [33], which rely on offline partitioning and joint tour optimization over a known grid model, the CFR-MARL offers a complementary learning-based solution that preserves high coverage efficiency and safety even when terrain layout, obstacles, and effective energy usage patterns must be inferred online from local, partial observations encoded in the shared DM.

Regarding the computational and communication overhead of the evaluated methods, the MADQN baseline incurs the lowest overhead, since each agent trains and executes its DQN independently using only local observations and the shared DM, with no requirement for global value computation or network parameter synchronization. The CFR-MARL introduces a moderate increase in computational cost, as each agent must additionally compute the local max-Q for the global potential function and participate in potential-based reward shaping; however, the required network parameter synchronizations are infrequent, so real-time communication demands remain low. In contrast, the VDN imposes the highest training-time burden but minimal execution-time overhead, as maintaining the accuracy of the centralized value decomposition necessitates frequent (often step-wise) parameter exchanges and coordinated Q-value updates across all agents, which can become a significant bottleneck in distributed or bandwidth-constrained deployments.

Finally, in degraded-link tests (5% packet loss, 0.5–1 s delay), the proposed CFR-MARL's performance degrades gracefully, coverage time increases by only ~7% and success rate drops <5%. This further underscores its resilience in complex environments such as Mars. It makes CFR-MARL an attractive middle ground, balancing the scalability of MADQN with the coordination benefits of VDN, yet avoiding the excessive synchronization costs of the latter.

4.6. Scalability test result

To evaluate how well the methods scale with increasing numbers of UGVs, additional simulations were conducted in the same 36×36 grid environment with 15% obstacles density, varying the UGV team size from 2 to 5 while retaining one UAV. Table 3 reports key performance metrics for the baseline VDN with DM method and the proposed CFR-MARL.

With each additional UGV, the CFR-MARL achieves a roughly linear reduction in completion time (≈ 540 s saved per additional UGV), whereas the baseline shows a slightly slower reduction (≈ 582 steps per UGV). By five UGVs, the CFR-MARL completes in 1670s (49% faster than two-UGV scenarios). As expected, overlap grows with more agents. However, the CFR-MARL maintains significantly lower redundancy, thanks to centralized feedback and shaping that coordinate agent division. Baseline reliability degrades (88.5%→70.2%) and collisions increase (9%→26%) as team size grows. In contrast, the CFR-MARL sustains high reliability (98.5%→95.3%) and low collisions (0.9%→3.6%), indicating robust coordination under scale.

The underlying reason is that in the pure VDN approach, agents share the global value function only during training. However, during execution, each agent relies solely on sparse intrinsic rewards. As a result, agents may occasionally revisit already-covered cells due to limited local observations. In contrast, under CFR-MARL, revisiting an

Table 3

Key performance metrics for the baseline VDN and the CFR-MARL method across varying UGV team sizes.

UGVs	VDN TC (s)	VDN OR (%)	VDN SR (%)	VDN CR (%)	CFR-MARL TC (s)	CFR-MARL OR (%)	CFR-MARL SR (%)	CFR-MARL CR (%)
2	3670	14.9	88.5	8.7	3280	4.8	98.5	0.9
3	2775	18	82.0	12.4	2400	5.9	97.9	1.7
4	2220	24.1	77.0	18.5	1920	8.03	96.7	2.5
5	1925	29.8	70.2	26.0	1670	11.5	95.3	3.6

already-covered cell never increases the global potential function, and thus always incurs a negative shaping bonus for the agent. Over time, this mechanism discourages behaviors that lead to redundant coverage, effectively driving the observed overlap ratio toward zero.

Furthermore, collisions are encoded as large negative intrinsic rewards within each Q_i . Consequently, any action sequence that brings agents into potential collision courses reduces the maximal sum of Q -values, and thus decreases potential function. The shaping term provides immediate negative feedback for collision-prone actions, offering much denser guidance than waiting for a sparse collision penalty. This dense, global feedback enables agents to proactively avoid each other and obstacles, resulting in a lower empirical collision rate.

These results demonstrate that the CFR-MARL not only outperforms classical methods in two-agent scenarios but also scales more efficiently and robustly as the team size increases. The superior scalability of CFR-MARL underscores its suitability for large-scale heterogeneous multi-agent deployments in resource-constrained environments such as Mars exploration.

Both algorithms were further evaluated by increasing the team to ten UGVs and two UAVs in the 36×36 grid with 15 % obstacle density. As shown in Fig. 6, the VDN with DM baseline required approximately 950 s to complete coverage, incurred 35 % overlap, achieved a 66 % success rate, and registered 27 % collision events. In contrast, the CFR-MARL completed the mission in 840 s, reduced overlap to 19 %, increased the success rate to 88 %, and lowered collisions to 9 %. These results demonstrate that incorporating potential-based reward shaping and UAV–UGV heterogeneity substantially improves coverage efficiency, mission reliability, and safety as team size increases.

5. Conclusions

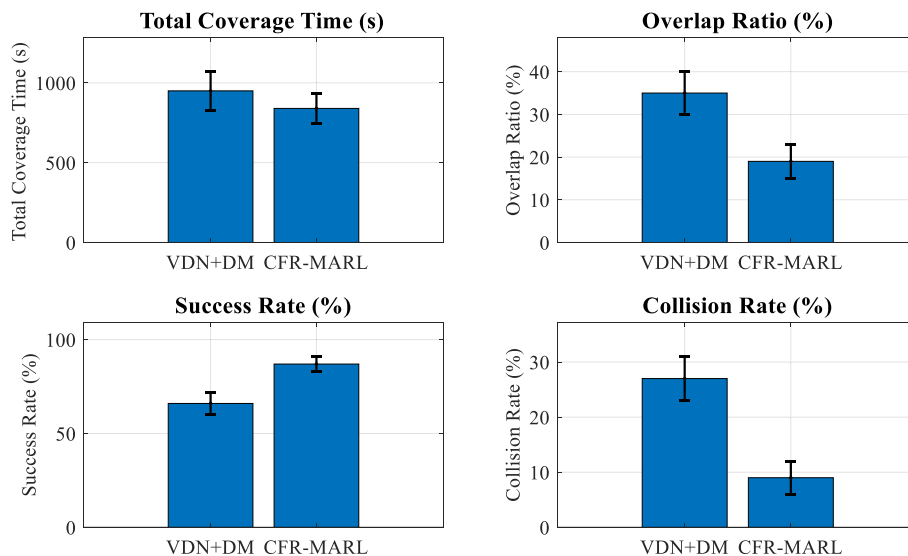
The proposed CFR-MARL framework demonstrated clear improvements in coverage efficiency and coordination for heterogeneous UAV–UGV teams operating in a Mars-like environment. Across all scenarios, CFR-MARL achieved faster mission completion times with less

redundant traversal than the considered baselines. The assisting UAV enabled the UGVs to follow shorter and better-coordinated paths, which reduced overlap and improved the balance of workload among ground agents. The shared DM supported effective communication between agents, allowing real-time adjustments of paths and a more consistent division of the area into subregions.

The UAV also managed its energy effectively. By learning when and on which UGV to recharge, it maintained continuous operation over long missions while still contributing meaningfully to coverage. The low collision rate observed in the experiments shows that the framework handled obstacle avoidance and coordination reliably, with the DM providing up-to-date information about discovered obstacles and covered cells.

However, the current study also has important limitations. The simulation assumes ideal communication, static obstacles and perfectly classified DM entries. Misclassifications of aerial imagery or ground sensing are not explicitly modelled, and the monotone DM update rule prevents obstacle labels from being revised within an episode, which may not fully reflect real-world Mars conditions. In addition, the evaluation is limited to high-fidelity simulation and does not include physical experiments with real UAV–UGV platforms.

Future work will focus on incorporating communication delays, dynamic obstacles, and testing the framework on physical platforms. Another promising direction is to replace the dense grid-based DM with a more compact sparse representation, for example a graph-based generative map in which new states and obstacle labels are instantiated and updated from agent observations in real time. Such a representation would allow false obstacle detections to be corrected when new evidence becomes available and would improve resilience to misclassified aerial imagery. The CFR-MARL would then operate on this learned latent environment model instead of a fixed dense array, preserving the same multi-agent RL structure while improving scalability in large or highly elongated Mars-like environments. Furthermore, the current study is limited to high-fidelity simulation, and no physical experiments are conducted. Real-world validation on UAV–UGV platforms,

**Fig. 6.** Metrics comparison for 10 UGVs and 2 UAVs.

particularly for obstacle detection using onboard UAV sensors, is an important direction for future work.

CRedit authorship contribution statement

M. Ramezani: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **M.A. Amiri Atashgah:** Supervision.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT as a grammar assistance tool. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] C. Dinelli, et al., Configurations and applications of multi-agent hybrid drone/unmanned ground vehicle for underground environments: a review, *Drones* 7 (2) (2023) 136.
- [2] B. Arbanas, A. Ivanovic, M. Car, M. Orsag, T. Petrovic, S. Bogdan, Decentralized planning and control for UAV–UGV cooperative teams, *Auton. Robots* 42 (8) (2018) 1601–1618.
- [3] C. Ersü, E. Petlenkov, K. Janson, A systematic review of cutting-edge radar technologies: applications for unmanned ground vehicles (UGVs), *Sensors* 24 (23) (2024) 7807.
- [4] D. Debnath, F. Vanegas, J. Sandino, A.F. Hawary, F. Gonzalez, A review of UAV path-planning algorithms and obstacle avoidance methods for remote sensing applications, *Remote Sens.* 16 (21) (2024) 4019.
- [5] K. Kumar, N. Kumar, Region coverage-aware path planning for unmanned aerial vehicles: a systematic review, *Physical Communication* 59 (2023) 102073.
- [6] S. Ramasamy, J.-P.F. Reddinger, J.M. Dotterweich, M.A. Childers, P.A. Bhounsule, Coordinated route planning of multiple fuel-constrained unmanned aerial systems with recharging on an unmanned ground vehicle for mission coverage, *J. Intell. Rob. Syst.* 106 (1) (2022) 30.
- [7] P. Fankhauser, et al., "Collaborative navigation for flying and walking robots," in: 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2016, pp. 2859–2866.
- [8] N. Khabbaz, S. Nokleby, UAV obstacle mapping for Multi-UGV exploration and mapping, in: Combined IFToMM Symposium of RoManSy and the USCToMM Symposium on Mechanical Systems and Robotics, Springer, 2024, pp. 155–166.
- [9] H.G. Changela, et al., Mars: new insights and unresolved questions, *Int. J. Astrobiol.* 20 (6) (2021) 394–426.
- [10] J. Shen, Y. Chen, Y. Sun, L. Liu, Y. Pan, W. Lin, Detection of biosignatures in Terrestrial analogs of Martian regions: strategical and technical assessments, *Earth Planet. Phys.* 6 (5) (2022) 431–450.
- [11] A. Banerjee, M. Mukherjee, S. Satpute, G. Nikolakopoulos, Resiliency in space autonomy: a review, *Current Robotics Reports* 4 (1) (2023) 1–12.
- [12] R. Chai, Y. Guo, Z. Zuo, K. Chen, H.-S. Shin, A. Tsourdos, Cooperative motion planning and control for aerial-ground autonomous systems: methods and applications, *Prog. Aero. Sci.* 146 (2024) 101005.
- [13] N.A. Olmedo, M. Barczyk, H. Zhang, W. Wilson, M.G. Lipsett, A UGV-based modular robotic manipulator for soil sampling and terramechanics investigations, *J. Unmanned Veh. Syst.* 8 (4) (2020) 364–381.
- [14] M. Ramezani, M.A. Alandihallaj, A.M. Hein, PPO-based dynamic control of uncertain floating platforms in Zero-G environment, in: 2024 IEEE International Conference on Robotics and Automation (ICRA), IEEE, 2024, pp. 11730–11736.
- [15] J. Cai, A. Du, I. Awolusi, Gaze-directed UAV-UGV coordination framework for onsite quality inspection of precast bridge construction, *Transportation Infrastructure Precast Innovation Center (TRANS-IPIC) Tier-1* (2024).
- [16] M. Ramezani, M. Atashgah, J.L. Sanchez-Lopez, H. Voos, Human-centric aware UAV trajectory planning in search and rescue missions employing multi-objective reinforcement learning with AHP and similarity-based experience replay, in: 2024 International Conference on Unmanned Aircraft Systems (ICUAS), IEEE, 2024, pp. 177–184.
- [17] M. Ramezani, M. Amiri Atashgah, A. Rezaee, A fault-tolerant multi-agent reinforcement learning framework for unmanned aerial vehicles–unmanned ground vehicle coverage path planning, *Drones* 8 (10) (2024) 537.
- [18] Z. Khanam, Coverage Path Planning for Autonomous Robots, University of Essex, 2022.
- [19] Q. Miao, G. Wei, A comprehensive review of path-planning algorithms for planetary rover exploration, *Remote Sens.* 17 (11) (2025) 1924.
- [20] K. Asadi, et al., An integrated UGV-UAV system for construction site data collection, *Autom. Construct.* 112 (2020) 103068.
- [21] Z.H. Ismail, N. Sariff, E.G. Hurtado, A survey and analysis of cooperative multi-agent robot systems: challenges and directions, *Applications of Mobile Robots* 5 (2018) 8–14.
- [22] M. Maimone, J. Biesiadecki, E. Tunstel, Y. Cheng, C. Leger, Surface navigation and mobility intelligence on the Mars Exploration Rovers, *Intelligence for space robotics* (2006) 45–69.
- [23] S. Farrell, C. Li, H. Yu, H. Mojtahedi, S. Gao, H.I. Christensen, "GLIDE: a Coordinated Aerial-Ground Framework for Search and Rescue in Unknown Environments," arXiv preprint arXiv:2509.14210 (2025).
- [24] M. Ramezani, M. Amiri Atashgah, Energy-aware hierarchical reinforcement learning based on the predictive energy consumption algorithm for search and rescue aerial robots in unknown environments, *Drones* 8 (7) (2024) 283.
- [25] N. Karapetyan, A.B. Asghar, A. Bhaskar, G. Shi, D. Manocha, P. Tokekar, Ag-cvg: coverage planning with a mobile recharging ugv and an energy-constrained uav, in: 2024 IEEE International Conference on Robotics and Automation (ICRA), IEEE, 2024, pp. 2617–2623.
- [26] A.B. Asghar, et al., Risk-aware recharging rendezvous for a collaborative team of uavs and uavs, in: Proceedings of the IEEE International Conference on Robotics and Automation, IEEE, 2023.
- [27] S. Brotee, et al., Optimizing UAV-UGV coalition operations: a hybrid clustering and multi-agent reinforcement learning approach for path planning in obstructed environment, *Ad Hoc Netw.* 160 (2024) 103519.
- [28] D.S. Drew, Multi-agent systems for search and rescue applications, *Current Robotics Reports* 2 (2021) 189–200.
- [29] L. Li, et al., Intelligent multi-robot exploration in non-exposed spaces: methods and challenges, *Artif. Intell. Rev.* 58 (12) (2025) 394.
- [30] G. Zijlstra, K.L. Aplin, E.R. Hunt, Multi-robot strategies for communication-constrained exploration and electrostatic anomaly characterization, in: 2024 International Conference on Space Robotics (iSpaRo), IEEE, 2024, pp. 76–83.
- [31] A. Jamshidepy, M. Wahby, M. Allwright, W. Zhu, M. Dorigo, M.K. Heinrich, Centralization vs. decentralization in multi-robot sweep coverage with ground robots and UAVs, *Artif. Life Robot.* (2025) 1–22.
- [32] G. Salavasisidis, et al., Terrain-aided navigation with coarse maps—Toward an arctic crossing with an AUV, *IEEE J. Ocean. Eng.* 46 (4) (2021) 1192–1212.
- [33] X. Lin, Y. Yazicioğlu, D. Aksaray, Robust planning for persistent surveillance with energy-constrained UAVs and mobile charging stations, *IEEE Rob. Autom. Lett.* 7 (2) (2022) 4157–4164.
- [34] G. Niu, L. Wu, Y. Gao, M.-O. Pun, Unmanned aerial vehicle (UAV)-assisted path planning for unmanned ground vehicles (UGVs) via disciplined convex-concave programming, *IEEE Trans. Veh. Technol.* 71 (7) (2022) 6996–7007.
- [35] Y. Wu, S. Wu, X. Hu, Cooperative path planning of UAVs & UGVs for a persistent surveillance task in urban environments, *IEEE Internet Things J.* 8 (6) (2020) 4906–4919.
- [36] P. A. Cook, Improved two-vehicle look-ahead Control Strategy.
- [37] K. Lochan, A. Khan, I. Elsayed, B. Suthar, L. Senneviratne, I. Hussain, Advancements in precision spraying of agricultural robots: a comprehensive review, *IEEE Access* (2024).
- [38] M. Saha and P. Isto, "Multi-robot motion planning by incremental coordination," in 2006 IEEE/RSJ International Conference on Intelligent Robots and Systems, 2006: IEEE, pp. 5960–5963.
- [39] B. Ying, Z. Su, Q. Xu, X. Ma, Game theoretical bandwidth allocation in UAV-UGV collaborative disaster relief networks, in: 2021 IEEE 23rd Int Conf on High Performance Computing & Communications; 7th Int Conf on Data Science & Systems; 19th Int Conf on Smart City; 7th Int Conf on Dependability in Sensor, Cloud & Big Data Systems & Application, HPCC/DSS/SmartCity/DependSys), IEEE, 2021, pp. 1498–1504.
- [40] O.S. Oubbati, J. Alotaibi, F. Alromithy, M. Atiquzzaman, M.R. Altmanian, A UAV-UGV cooperative system: patrolling and energy management for urban monitoring, *IEEE Trans. Veh. Technol.* (2025).
- [41] Y. Huang, S. Wu, Z. Mu, X. Long, S. Chu, G. Zhao, A multi-agent reinforcement learning method for swarm robots in space collaborative exploration, in: 2020 6th International Conference on Control, Automation and Robotics (ICCAR), IEEE, 2020, pp. 139–144.
- [42] E.A. Abaku, T.E. Edunjobi, A.C. Odimarha, Theoretical approaches to AI in supply chain optimization: pathways to efficiency and resilience, *International Journal of Science and Technology Research Archive* 6 (1) (2024) 92–107.
- [43] S. Khoroshlyov, M. Redka, Deep learning for space guidance, navigation, and control, *Space Science and Technology* 27 (6) (2021) 38–52.
- [44] C. Mu, S. Liu, M. Lu, Z. Liu, L. Cui, K. Wang, Autonomous spacecraft collision avoidance with a variable number of space debris based on safe reinforcement learning, *Aero. Sci. Technol.* 149 (2024) 109131.
- [45] M. Ramezani, M. Alandihallaj, A.M. Hein, Fuel-efficient and fault-tolerant CubeSat orbit correction via machine learning-based adaptive control, *Aerospace* 11 (10) (2024).
- [46] M. Ramezani, M.A. Alandihallaj, B.C. Yalçın, M.A.O. Mendez, H. Voos, "MPC-based Deep Reinforcement Learning Method for Space Robotic Control with Fuel Sloshing Mitigation," arXiv preprint arXiv:2509.21045 (2025).
- [47] M. Ramezani, M.A. Alandihallaj, B.C. Yalçın, M.A.O. Mendez, A.M. Hein, Fuel-aware autonomous docking using RL-augmented MPC rewards for On-Orbit refueling, *Acta Astronaut.* (2025).

- [48] M. Ramezani, M. Atashgah, M. Alandihallaj, A. Hein, Reinforcement Learning for planning and task coordination in a swarm of CubeSats: overcoming processor limitation challenges, in: International Astronautical Congress, 2023.
- [49] M. Ramezani, M.A. Alandihallaj, J.L. Sanchez-Lopez, A. Hein, "Safe Hierarchical Reinforcement Learning for CubeSat Task Scheduling Based on Energy Consumption," arXiv preprint arXiv:2309.12004 (2023).
- [50] M.A. Alandihallaj, M. Ramezani, A.M. Hein, MBSE-enhanced LSTM framework for satellite system reliability and failure prediction, MODELSWARD (2024) 349–356.
- [51] M. Ramezani, H. Habibi, J.L. Sanchez-Lopez, H. Voos, UAV path planning employing MPC-reinforcement learning method considering collision avoidance, in: 2023 International Conference on Unmanned Aircraft Systems (ICUAS), IEEE, 2023, pp. 507–514.
- [52] G. Shetty, M. Ramezani, H. Habibi, H. Voos, J.L. Sanchez-Lopez, Motion control in multi-rotor aerial robots using deep reinforcement learning, in: 2025 International Conference on Unmanned Aircraft Systems (ICUAS), IEEE, 2025, pp. 29–36.
- [53] M. Ramezani, H. Habibi, H. Voos, UAV path planning employing MPC-reinforcement learning method considering collision avoidance, arXiv preprint arXiv:2302.10669 (2023).
- [54] D. Jaganathan, V.K. Kaliappan, Optimizing urban sustainability: reinforcement learning-driven energy-efficient ubiquitous robots for smart cities, in: Intelligent Solutions for Sustainable Power Grids, IGI Global, 2024, pp. 245–271.
- [55] Y. Huang, "Deep Q-networks," Deep Reinforcement Learning: Fundamentals, Research and Applications, 2020, pp. 135–160.
- [56] T. Ding, L. Liu, Z. Yan, L. Cui, Enhancing sustainable data collection with unmanned aerial vehicles: a deep reinforcement learning approach, IEEE Internet Things J. (2024).
- [57] K. Zhang, Z. Yang, T. Başar, Multi-agent reinforcement learning: a selective overview of theories and algorithms. Handbook of Reinforcement Learning and Control, 2021, pp. 321–384.
- [58] A. Barth, O. Ma, Cooperative behavior of a heterogeneous robot team for planetary exploration using deep reinforcement learning, Acta Astronaut. 214 (2024) 689–700.
- [59] M.P. Deisenroth, G. Neumann, J. Peters, A survey on policy search for robotics, Foundations and Trends® in Robotics 2 (1–2) (2013) 1–142.
- [60] M. Hua, et al., Multi-agent reinforcement learning for connected and automated vehicles control: recent advancements and future prospects, IEEE Trans. Autom. Sci. Eng. (2025).
- [61] T. Rashid, M. Samvelyan, C.S. De Witt, G. Farquhar, J. Foerster, S. Whiteson, Monotonic value function factorisation for deep multi-agent reinforcement learning, J. Mach. Learn. Res. 21 (178) (2020) 1–51.
- [62] P. Qin, S. Wang, Z. Lu, Y. Xie, X. Zhao, Deep reinforcement learning-based energy minimization task offloading and resource allocation for air ground integrated heterogeneous networks, IEEE Syst. J. 17 (3) (2023) 4958–4968.
- [63] P. Sunehag, et al., "Value-decomposition networks for cooperative multi-agent learning," arXiv preprint arXiv:1706.05296 (2017).
- [64] T.T. Nguyen, N.D. Nguyen, S. Nahavandi, Deep reinforcement learning for multiagent systems: a review of challenges, solutions, and applications, IEEE Trans. Cybern. 50 (9) (2020) 3826–3839.
- [65] G. He, S. Cui, Y. Dai, T. Jiang, Learning task-oriented channel allocation for multi-agent communication, IEEE Trans. Veh. Technol. 71 (11) (2022) 12016–12029.
- [66] M.L. Puterman, Markov decision processes, Handb. Oper. Res. Manag. Sci. 2 (1990) 331–434.
- [67] A.Y. Ng, D. Harada, S. Russell, Policy invariance under reward transformations: theory and application to reward shaping, Icml 99 (1999) 278–287. Citeseer.